# Venkata Lakshmi Parimala Pasupuleti

Agentic AI Engineer · LangGraph Multi-Agent · HITL Orchestration · LLM Evaluation · FHIR · HIPAA · RCM · Python · Durable State · Production AI

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venkatalakshmiparimala.github.io/Portfolio | github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

## PROFESSIONAL SUMMARY

Senior Software Engineer with 8 years of production systems experience and deep focus on agentic AI orchestration, LLM evaluation, and healthcare systems. Built **Leasel** - an 8-agent LangGraph StateGraph orchestration platform with durable state management, conditional HITL escalation, citation-grounded RAG (hybrid TF-IDF + semantic reranking), and a deterministic/probabilistic blend where LLM extracts and reasons while deterministic logic makes all regulated decisions. The exact architecture Maestro is building. Built **AgentProbe** - an open-source eval-driven development toolkit with a CI-integrated 5-metric regression harness, golden datasets, LLM-as-judge scoring, and offline + online evaluation loops - the evaluation infrastructure needed to ship non-deterministic systems safely to healthcare customers. Built **AI on FHIR** - FHIR APIs + HIPAA-compliant RBAC + NLP-driven clinical query system, satisfying the HL7/FHIR and PHI compliance required qualifications directly. At Entrata (8yr): multi-tenant production systems for 100K+ housing units, 100% PHI compliance on AWS S3 migration with Libsodium-encrypted pipelines, RabbitMQ idempotency + retry guarantees eliminating double-processing, and **Revenue Cycle Management (RCM)** domain expertise. Python expert, C# 1.5yr, TypeScript. Temporal, AWS Step Functions, Kubernetes.

## CORE COMPETENCIES

Agentic Orchestration | Multi-Agent Architectures | LangGraph / LangChain | HITL / Human-in-the-Loop | Durable Workflows | Long-Running Execution | Idempotency & Replayability | State Management | LLMs / Tool-Calling / Structured Outputs | Eval-Driven Development | LLM Regression Testing | Non-Deterministic Systems | RAG / Hybrid Search / Citation Grounding | HL7 / FHIR | HIPAA / PHI Compliance | Revenue Cycle Management (RCM) | Multi-Tenant Systems | Temporal | AWS Step Functions | Kubernetes / Docker | Python | C# | TypeScript / React | FastAPI / REST APIs | CI/CD | Prototype to Production | Distributed Systems | RabbitMQ / Celery / Kafka | PostgreSQL / Redis | AWS Bedrock / SageMaker

## EXPERIENCE

### Blue Horse Digital

Sep 2026 - Present

**AI/ML Engineer (Volunteer) — Virginia, USA**

- Developing ML models and data analysis pipelines for software applications and business intelligence; integrating data systems and AI components with backend/frontend teams; building pipelines that surface insights from raw business data into application-facing APIs.

### Magizhs Tech Solution

May 2025 - Aug 2025

**AI/ML Engineer Intern**

- Built an **agentic mission-planning system** orchestrated as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) with durable state - risk-aware compliance strategy routes compliant missions to export, non-compliant to rejection with grounded reasons.
- Implemented a **RAG layer** (retrieval over FAA Part 107 + operational-guidance corpus) answering operator queries with cited sources; engineered validation guardrails enforcing FAA altitude limits, battery budgets, and geometric no-fly-zone checks - blocking non-compliant missions before export.

### George Mason University

Aug 2024 - Aug 2025

**Research Associate / Graduate Teaching Assistant — Virginia, USA**

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets across 190+ countries - $R^2$=0.977 regression accuracy; applied Isolation Forest anomaly detection across 3 research pipelines.
- Fine-tuned a custom classification model using HuggingFace Transformers and PyTorch; deployed inference API on Quart (async Python web framework); presented model design, training methodology, and evaluation results to faculty.
- Delivered technical workshops to 600+ students across multiple sections of MIS 303 covering Python programming and database design.

### Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

**Senior Software Engineer — Lehi, UT (Remote)**

- **Owned the annual HUD income-limit import pipeline for nearly 7 years** - few thousand federal source rows expanded into ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units; zero compliance violations across ownership period.
- **Built and maintained the RabbitMQ-based async processing system** for gross-rent updates - designed **retry logic and idempotency guarantees** that eliminated double-processing failures; exact durable-state + replayability pattern required for long-running agentic workflows.
- **Led AWS S3 document migration with 100% PHI compliance** - Libsodium-encrypted pipelines across all resident data; zero security incidents; HIPAA-compliant multi-tenant data isolation for 100K+ monthly active users.
- **Optimized 200+ SQL queries (30% performance gain)**; owned complex multi-tenant schemas for eligibility, financial, and document workflows; shipped 20+ full-stack production modules across 120+ Agile sprints; **Revenue Cycle Management (RCM)** domain ownership.
- Led microservices redesign using Docker - 50% scalability improvement, 30% deployment time reduction; mentored 5+ junior engineers; conducted 300+ peer code reviews.

### Entrata India - Db Xento Systems

Jan 2017 - Sep 2018

**Associate / Trainee Software Engineer — Pune, India**

- Built property management modules (PHP, Python, Java, PostgreSQL) for affordable housing including income limits import (1.6M HUD/Tax Credit records across 5 years), document management, and compliance workflows; led cloud migration of client documents to AWS S3.

## PROJECTS

### Leasel: Agentic AI Support Investigation Platform

LangGraph · Multi-Agent · HITL · RAG · Durable State

Built a three-surface enterprise AI knowledge platform on a single evidence model: (1) **8-agent LangGraph StateGraph** with durable state (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) - **conditional HITL escalation** when confidence is too low for automated decision; (2) **Hybrid RAG** - TF-IDF + semantic reranking, citation grounding to exact source clauses, missing/conflicting clause detection; (3) Core design rule enforced in code: **LLM extracts and reasons; deterministic logic makes the final regulated call** - exact deterministic/probabilistic blend Maestro requires for healthcare workflows. Achieved 93.6% precision and 87.9% F1 across 10 compliance rules on 1,240 residential lease PDFs; pass@1 81.7% via regression gate + LLM-as-judge.

Python · LangGraph (8-agent) · FastAPI · TF-IDF · Semantic Reranking · OCR · SQLite · ChromaDB · 2026

### AgentProbe: Open-Source AI Evaluation Toolkit

Eval-Driven Dev · LLM Regression · CI/CD

Built a fully local, zero-cost **AI agent evaluation toolkit** for non-deterministic systems: 15-endpoint FastAPI backend, **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate), **CI-integrated regression harness** with golden datasets and LLM-as-judge scoring, model routing across OpenAI/Anthropic/Groq/Ollama, and **offline + online evaluation loops**. Used to regression-test Leasel across 1,240+ compliance test cases - the evaluation infrastructure needed to ship LLM-powered systems safely in production healthcare environments.

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · 2025 · github.com/venkatalakshmiparimala/AgentProbe

### AI on FHIR: NLP-Driven Medical Records Query System

FHIR · HIPAA · Clinical NLP · RCM

Reduced manual clinical retrieval time ~60% by building a Flask + React NLP platform translating natural-language queries (EN/ES) into **FHIR API calls** using LLMs, BERT, and embeddings. Enforced **HIPAA-compliant RBAC** across structured medical records - PHI access controlled at the data layer. Supports clinical workflow queries including **Revenue Cycle Management (RCM)** record retrieval. Containerized with Docker; multilingual clinical NLP using spaCy + BERT.

Flask · React · FHIR APIs · HIPAA RBAC · LLMs · BERT · spaCy · Embeddings · Docker · 2024

## EDUCATION

George Mason University **- M.S. Data Analytics Engineering** May 2026

GPA: 3.87 · Virginia, USA · Coursework: ML systems, NLP, cloud infrastructure, statistical modeling, deep learning

Jawaharlal Nehru Technological University **- B.Tech Computer Science & Engineering** 2017